

Module - 4 - Property, Plant and Equipment.

Introduction :-

The objectives of Indian Accounting Standard 16. The Accounting treatment for Property, Plant and Equipment. So that users of the financial Statement can find out information about the Company investment in its ~~prop~~ Property, Plant and Equipment and these changes in such Investment.

Main Issued :-

The Recognition of the assets. The determination of their carrying amounts. The depreciation charges, Impairment losses to be Recognised in relation to them.

Meaning of Impairment losses :-

An Impairment loss is the amount by which an asset's recorded value exceed that can be realized recovered from selling it.

Scope :-

1) Definition :- Property, Plant and Equipment are tangible ^{asset} items that :

a) Are held for use in the production or supply of goods or services² for rental to others or for administrative purpose.

b) Are expected to be used during more than one Period.

2) PPE Accounting :- This Standard shall be applied in accounting for Property, Plant and Equipment ~~ext~~ except when ~~Another~~ another Standard requires or permits a different accounting treatment.

3) Exception :- This Standard does not apply to :

- a) Property, Plant and Equipment classified as held for sale.
- b) Recognition and measurement of Exploration and Evaluation assets.
- c) Biological assets related agricultural activity.
- d) Mineral rights and mineral reserves such as oil, natural gas and similar Non-Regenerative resources.

4) Investment Property :- An entity Accounting for investment property in accordance with Ind AS 40 'Investment Property' shall use the Cost model in this Standard.

* Property and Plant :

1) Condition :- The Cost of an item of an property, plant and Equipment shall be recognised as an assets if an only if ::

a) It is probable that future Economic benefits associated with the items will follow to the entity and.

by The Cost of item can be measured reliably.

2) Initial and Subsequent Costs : An entity evaluates this recognition principle all its property, plant and Equipment Costs at the time they are incurred.

a) Costs incurred initially to acquire or construct an item of property, plant and equipment and.

b) Costs incurred ~~sub~~ subsequently to add, to replace part, or services it or major overhaul inspection.

3) Spares :- Items such as spares part, stand by equipment and services equipment are recognised in accordance with this Ind AS when they meet the definition of property, plant and equipment. otherwise, such as items are classified as inventory.

Initial measurement :- Initial measurement of property, plant and equipment is carried at cost items of property, plant and equipment may be acquired for safety or environmental reason. The acquisition of such property, plant and equipment although not directly increasing the ~~for~~ future economic benefits of any particular existing item of property plant and equipment, may be necessary for an entity to obtain the future economic benefits from its other assets. Such items of property, plant and equipment qualify for recognition as assets because they enable an entity to derive future economic benefits from related assets in excess of what could be derived had those items not been acquired.

For e.g.:- A Chemical manufacture may install new Chemical handling processes to comply with environmental requirements for the production and Storage of dangerous Chemicals.

4.6. Subsequent Measurement :- Cost model versus revaluation model :- Subsequent Costs are recognised in profit or loss as incurred. The purpose of these expenditure is often described as 'for the Repairs and maintenance' of the item of property, plant and Equipment. Parts of some items of Property, Plant and Equipments may require replacement at regular intervals. for example, A furnace may require relining after a specified number of hours of use, or Aircraft interior such as seats and galleys may required replacement several times during the life of the airframe.

1) The replacement will bring future benefits (longer life, better use).

2) The cost can be measured reliably.

At the same time, the old part should be removed (derecognized) Routine repairs like repainting or fixing cracks are not capitalized. They are recorded as expenses right away.

1) It gives future benefits.

2) The cost can be measured reliably.

Cost Incurred on routine checks are expensed.

Subsequent Expenditure

Regular / day-to-day
Repairs and
maintenance

Expensed to P/L

Expenditure which
increase life or
efficiency of the
asset beyond the
originally assessed
life or efficiency.

Capitalised

Major replacement /
major inspection /
major overhaul

Capitalised as
below:

$$\text{WDV of Asset} + \text{Cost of new part} - \text{WDV of old part} = \text{Revised WDV.}$$

* We will see the difference between Initial measurement and the subsequent measurement

Aspect	Initial Measurement	Subsequent Measurement
1. Basis of Measurement	Initial measurement of PPE is always at Cost.	for Subsequent measurement of PPE, there is a choice between Cost model or Revaluation model.
2. Included Components.	Purchase Price, import duties, non-refundable taxes, directly attributable depreciation, impairment	Subsequent measurements and adjusted for

	Costs, dismantling / restoration obligations are included in cost.	losses, or revaluation adjustments.
3. Excluded Costs	Administrative overheads, Start-up Costs, abnormal wastage, training Costs are excluded from cost during initial measurement.	Administrative overheads, abnormal Costs etc. are already excluded at initial stage. So nothing is excluded at subsequent measurement.
4. Flexibility.	There is no choice in case of initial measurement. It must be cost.	Entity can choose cost model or revaluation model.
5. Impact on Equity.	There is no impact on equity.	Revaluation Surplus. Credited to OCI and accumulated in equity.
6. Impact on Profit & Loss.	There is no impact at recognition stage on profit & loss.	Depreciation, impairment losses, and revaluation decreases affect Profit & Loss.
7. Disclosure Requirement	Cost Components and recognition principles are disclosed.	Policy Choice, revaluation details, depreciation method, Carrying amounts are disclosed.

Illustration - 1: (Cost of Remodelling).

Entity A, a Supermarket Chain, is renovating one of its major stores. The store will have more available space for in store promotion outlets after the renovation and will include a restaurant. Management is preparing the budgets for the year after the store reopens, which include the cost of remodeling and the expectation of a 15% increase in sales resulting from the store renovations, which will attract new customers. State whether the remodeling cost will be capitalized or not.

→ The expenditure in remodeling the store will create future economic benefits in the form of increase in sales and cost of remodeling can be measured reliably, Therefore it should be capitalized.

Illustration - 2: (Incremental Cost during Redevelopment).

Entity A has an existing freehold factory property, which it intends to knock down and redevelop.

During the redevelopment period the company will move its production facilities to another (temporary) site. The following incremental costs will be incurred:

1. Setup Costs of ₹ 5,00,000 to install machinery in the new location.
2. Rent of ₹ 15,00,000
3. Removal Costs of ₹ 3,00,000 to transport the machinery from the old location to the temporary location.

Can these Costs be Capitalised into the Cost of the new building ?

→ Constructing or acquiring a new assets may result in incremental Costs that would have been Avoided. If the assets had not been Constructed or acquired. These Costs are not to be included in Cost of the assets if they are not directly attributable to bringing the asset to the location and Condition necessary for it to be Capable of operating in the manner intended.

Illustration - 3: (Trial Runs).

An amusement park has a 'Soft opening' to the Public, to trial run its attractions. Tickets are sold at a 50% discount during this period and the operating Capacity is 80%. The official opening day of the amusement park is three months later. management Claims that the soft opening is a trial run necessary for the amusement park to be in the Condition Capable of operating in the intended manner. Accordingly, the net operating Costs incurred should be Capitalised.

Comment:

→ The net operating Costs incurred during the trial run should not be Capitalised. They should be Charged to the Statements of Profits and loss.

Although the amusement park was operating at 80% of its Capacity, there is sufficient evidence that

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the park was capable of operating in the manner intended by management. Therefore, the costs incurred during the trial run are operating expenses.

4.5. Initial Measurement:

Initial measurement of property, plant and equipment is carried at cost.

Items of property, plant and equipment may be acquired for safety or environmental reasons.

The acquisition of such property, plant and equipment although not directly increasing the future economic benefits of any particular existing item of property, plant and equipment, may be necessary for an entity to obtain the future economic benefits from its other assets. Such items of property, plant and equipment qualify for recognition as assets because they enable an entity to derive future economic benefits from related assets in excess of what could be derived had those items not been acquired.

for example, a chemical manufacturer may install new chemical handling processes to comply with environmental requirements for the production and storage of dangerous chemicals. These related plant enhancements are recognised as an asset because without them the entity is unable to manufacture and ~~sell~~ sell chemicals.

Measurement at Cost on Recognition :

An item of Property, Plant and Equipment that qualifies for recognition as an asset shall be measured at its cost.

Elements of Cost :

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Purchase Price, including import duties and non-refundable

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Purchase Price, including import duties and non-refundable purchase taxes, after deducting trade discounts and rebates.	Any Costs directly attributable to bringing the asset to the location and condition necessary.	Cost of dismantling and restoring the site later.
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Examples of Directly attributable costs:

1. Costs of employee benefits arising directly from the construction or acquisition of the item of property, plant and equipment.
2. Cost of site preparation.
3. Cost of initial delivery and handling costs.
4. Installation and assembly costs.
5. Cost of testing whether the asset is functioning properly, after deducting the net proceeds from selling any items produced while bringing the asset to that location and condition, (such as samples produced when testing equipment).
6. Professional fees.

* If an asset is purchased by payment for cash or for credit, or it is self constructed the cost of the asset includes the following:

Particulars	Amt (₹)
Purchase Price (Basic Price)	xxx
Add: Non-refundable taxes and import duties.	xxx
Add: Initial delivery and handling costs.	xxx
Add: Cost of employee benefits arising from the construction or acquisition of PPE.	xxx
Add: Site preparation cost.	xxx
Add: Installation and assembly cost.	xxx
Add: Cost of testing whether the asset is functioning properly.	xxx
Add: Professional fees	xxx
Less: Trade discount and rebates Cost of PPE to cap capitalised.	(xxx)
	xxx

Carrying Amount of an asset is calculated as given below:

Particulars	Amt. (₹)
Gross Book Value	₹xxx
Less: Accumulated Depreciation	(xxx)
Less: Accumulated impairment loss (if any)	(xxx)
Carrying Amount	<u>₹xxx</u>

* Examples of Costs that are not Cost of an item of Property, Plant and Equipment:

1. Cost of opening a new facility.
2. Cost of introducing a new product or service, (including the cost of advertising and promotional activities).
3. Cost of conducting business in a new location or with a new class of customers (including cost of staff training).
4. Administration and other general overhead costs.

* Examples of Costs Not included in the Carrying amount of an item of Property, Plant and Equipment:

1. Costs incurred while an item capable of operating in the manner intended by management has yet to be brought into use or is operated at less than full capacity.
2. Initial operating losses.
3. Cost of relocating or reorganising part or all of an entity's operations.

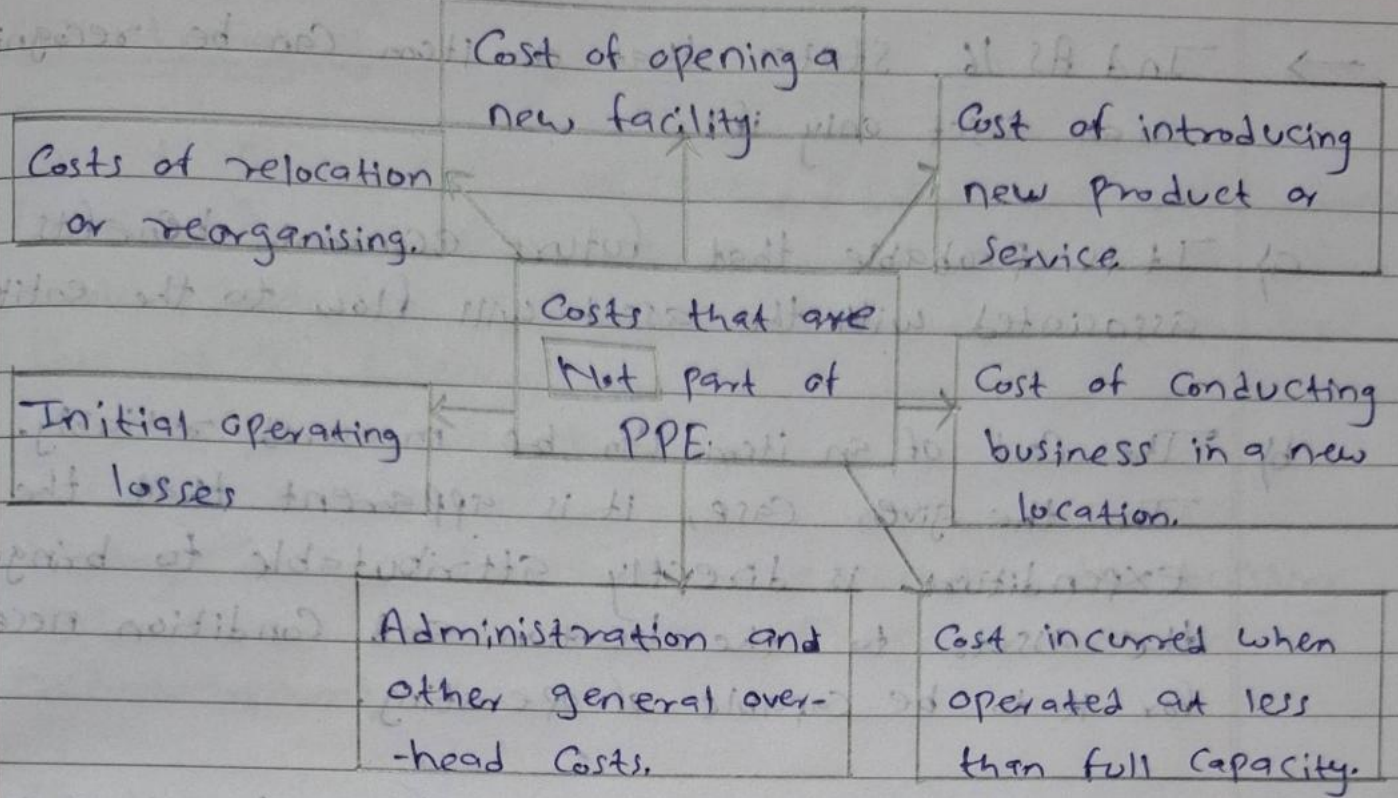


Illustration - 4 - (Railway Siding, Road and bridge).

ABC Ltd. is setting up a new refinery outside the city limits. In order to facilitate the construction of the refinery and its operations, ABC Ltd. is required to incur expenditure on the construction / development of railway siding, road and bridge. Though ABC Ltd. incurs (or contributes to) the expenditure on the construction / development, it will not have ownership rights on these items and they are also available for use to other entities and public at large.

Whether ABC Ltd. can capitalise expenditure incurred on these items as property, plant and equipment (PPE)? If yes, how should these items be depreciated and presented in the financial statements of ABC Ltd. as per Ind AS?

→ Ind AS 16 states that an item can be recognised as an asset only if:

a) It is probable that future economic benefits associated with the asset will flow to the entity.

b) The cost of an item can be measured reliably.
In the given case, it is apparent that the expenditure is directly attributable to bringing the assets to the location and condition necessary for it to be capable of operating.

Hence, the expenditure incurred will be capitalised as a part of the overall cost of the project.

From this it can be concluded that the expenditure incurred on this asset should be considered as the cost of constructing the refinery and accordingly expenses should be allocated and capitalised as part of the items of property, plant and ~~Equip~~ Equipment (PPE) of the refinery.

Illustration - 5.

On 1 April 2025, an entity purchased an office block (building) for ₹ 50,00,000 and paid a non-refundable property transferred tax and direct legal cost of ₹ 2,50,000 and ₹ 50,000 ~~Respect~~ respectively while acquiring the property.

During 2025, The entity redeveloped the building into two storey building. Expenditures on redevelopment were:

1. ₹ 1,00,000 building plan approval.
2. ₹ 10,00,000 Consulting Cost. (including ₹ 60,000 refundable purchase taxes).
3. ₹ 40,000 due to abnormal wastage of material and labour.

How will the entity account for all the above mentioned expenses in the books of Accounts?

→ In accordance with Ind AS - 16, All Cost required to bring an asset to its present location and condition for its use should be Capitalised. Therefore, The initial price of the building would be.

Particulars	Am't (₹)
Purchase Prices.	50,00,000
Add: Non-Refundable property taxes.	2,50,000
Add: Direct legal Cost	50,000
	<u>53,00,000</u>
Expenses on redevelopment	
1. Building plan approval	1,00,000
2. Construction Costs (10,00,000 - 60,000)	9,40,000
3. Abnormal wastage on material and labour.	-
Total amount to be Capitalised	<u>63,40,000</u>

Treatment of abnormal wastage of material and labour.

As per Ind AS - 16 The Cost of abnormal wastage is not included in the Cost of assets it will be charged to profit and loss A/c in the year it is incurred.

Illustration - 6 - (Measurement at initial Recognition).

On 1 January 2025 an entity purchased an item of equipment for ₹ 6,00,000 including ₹ 50,000 refundable purchase taxes.

In January 2025 the entity incurred costs of ₹ 20,000 in transporting the equipment to the entity's site and ₹ 1,00,000 in installing the equipment at the site. At the end of the equipment's 10-years useful life the entity is required to dismantle the equipment and restore the land upon which the factory is built.

The present value of the costs of dismantling the equipment and restoring the ~~environment~~ environment is estimated to be ₹ 1,00,000.

In January 2025 the entity's engineer incurred the following costs in modifying the equipment so that it can produce the products manufactured by the entity:

1. Material - ₹ 55,000
2. Labour - ₹ 65,000
3. Depreciation of plant and equipment used to perform the modifications - ₹ 15,000

In January 2025 the entity's production staff were trained in how to operate the new item of equipment.

Training costs included:

1. Cost of an expert external instructor - ₹ 7,000
2. Labour - ₹ 3,000

In February 2025 the entity's production team tested the equipment and the engineering team made further

modifications necessary to get the equipment to function as intended by management.

The following costs were incurred in the testing phase:

1. Material, net of ₹ 3,000 recovered from the sale of the scrapped output - ₹ 21,000.
2. Labour - ₹ 11,000.
3. Depreciation of plant and equipment used to perform the modifications - ₹ 5,000.

The equipment was ready for use on 1 March 2025. However, because of low initial order levels the entity incurred a loss of ₹ 23,000 on operating the equipment during March. Thereafter the equipment operated profitably.

What is the cost of the equipment at initial recognition?

→ Calculation of cost of equipment.

Particulars	Amount (₹)
Purchase Price (6,00,000 - 50,000 refundable taxes)	5,50,000
Add: Transport Cost (directly attributable cost)	20,000
Add: Installation Cost (directly attributable cost)	1,00,000
Add: Environmental restoration cost (cost of dismantling and restoring)	1,00,000
Add: Preparation cost (55,000 + 65,000 + 15,000)	1,35,000
Add: Training cost (Recognised as an expense in P&L A/c)	-
Add: Cost of testing (21,000 material + 11,000 Labour + 5,000 Dep.)	37,000
Add: operating loss (recognised as an expense in P&L A/c)	-
Cost of Equipment	9,42,000

Illustration - 7 (Depreciation Charge allocated to the Cost of self-constructed asset).

On 1 January 2025 an Entity acquired a machine for ₹ 12,00,000. Management estimates the machine has a 10-year useful life (measured from the date of acquisition) and a nil residual value. Furthermore, management judges that the Straight-line method reflects the pattern in which the entity ~~expense~~ expects to consume the machine's future economic benefits.

In 2025 the machine was used to produce inventory for eight months. Thereafter, the machine was used to manufacture components of a new item of plant being constructed by the entity. The new plant will be used by the entity to manufacture a new product line.

→ Formula : Depreciation for the year

$$= \frac{\text{Cost of asset} - \text{Scrap Value}}{\text{Useful life in years}}$$

$$= \frac{12,00,000 - \text{Nil}}{10}$$

$$\text{Dep.} = ₹ 1,20,000$$

In 2025 the entity must allocate ₹ 80,000 that is $1,20,000 \times \frac{8}{12}$

To the Cost of inventory and ₹ 40,000 to the Cost of new plant undergoing construction (that is $1,20,000 \times \frac{4}{12}$).

Illustration - 8 - (Asset acquired in Exchange).



Cost of assets received = Carrying Amount of assets given up - Cash received

$$95,000 = 1,00,000 - 5,000$$

Particulars	Debit (₹)	Credit (₹)
1. Asset A/c Dr	95,000	
Cash A/c Dr	5,000	
To Asset A/c		1,00,000

Illustration - 9 - (Asset Acquired in Exchange).

→ Car X : Book value (Carrying Amount) = ₹ 13,000
 Fair value = ₹ 13,250
 Exchange for Car Y, whose fair value is ₹ 13,100 also received Cash amounting to ₹ 150.

Since the transaction lacks Commercial Substance, the Company recognised the assets received at book value.

Therefore, the entry will be

Particulars		Debit (₹)	Credit (₹)
1. Asset A/c	Dr	12,850	
Cash A/c	Dr	150	
To Asset A/c			13,000

Illustration - 10. (Measurement After Revaluation).

→ Calculation of Depreciation for the ^{After} ~~5th~~ Year.
 Depreciation per Year charged for four years:

$$\text{Depreciation} = \frac{\text{Cost} - \text{Scrap Value}}{\text{Life in Year}}$$

$$= \frac{10,00,000 - \text{Nil}}{10}$$

$$= ₹ 1,00,000 \text{ per Year}$$

$$\therefore \text{Total Depreciation for 4. Years} = 1,00,000 \times 4 = 4,00,000$$

$$\text{Written-down Value of machine} = 10,00,000 - 4,00,000 = ₹ 6,00,000$$

$$\therefore \text{Dep. amount after revaluation} = 6,00,000 + 40,000 = 6,40,000$$

Remaining useful life as per revised estimate 6 Years (10-4)

Remaining useful life as per revised estimate = 8 Years.

$$\therefore \text{Depreciation for the fifth Year} = \frac{6,40,000 - \text{Nil}}{8} = ₹ 80,000$$

Illustration - 11 - (Useful life of an Asset).

As part of their remuneration package an entity provides each senior manager with the private use of a luxury motor vehicle of the manager's choice. The executive motor vehicles are replaced every two years irrespective of usage. The entity sells and replaces its luxury motor vehicle fleet every two years when the vehicles are expected to be economically usable by one or more users for at least another three years. Determine the useful life of the asset.

→ The useful life of the vehicles is two years.

Illustration - 12 - (Useful life of an Asset).

An entity does not service its equipment regularly. With regular servicing the equipment would be available for use for five years. However, the expected equipment servicing pattern is expected to render the equipment unusable in three years. Determine the useful life of the asset.

→ The useful life of the equipment is ~~three~~ three years.

Illustration - 13 - (Useful life of an asset).

An entity's equipment used to manufacture a patented drug is expected to be capable of producing the drug for ~~the~~ ten years. However, the entity expects to stop manufacturing the drug and scrap the equipment after five years of production when its patent expires and low cost generic drugs are expected to render the entity's manufacturing of this drug unprofitable. Determine the useful life of the asset.

→ The useful life of the equipment is five years.

Illustration - 14 - (Useful life of an Asset).

An entity has the right of use of an item of equipment in accordance with the terms of a finance lease. The equipment is capable of operating for 15 years. However, the lease term is 13 years and the entity is required to return the equipment to the lessor at the end of the lease term. ~~Determine~~ Determine the useful life of the asset.

→ The useful life of the equipment is 13 years.

Illustration - 16 - (Revised Assessment of useful life and Depreciation).

→ 1. Depreciation for Year Ended 2022, 2023 and 2024.

Particulars	Original Cost	Estimated Life	Dep. P.a.
Property, plant and Equipment:			
Building	1,50,00,000	15 Years	10,00,000
Plant and Machinery	1,00,00,000	10 Years	10,00,000
Furniture and Fixture	35,00,000	7 Years	5,00,000
Total Depreciation.			25,00,000

2. Depreciation for Year Ended 2025.

Particulars	Original Cost	Dep. (3 Years)	Carrying (1-4-2024)	Revised life	Dep.
Property, plant and Equipment:					
Building	1,50,00,000	30,00,000	1,20,00,000	10 Years	12,00,000
Plant	1,00,00,000	30,00,000	70,00,000	7 Years	10,00,000
Furniture	35,00,000	15,00,000	20,00,000	5 Years	4,00,000
Total					26,00,000